

Mechanics WS 8

1. A particle of mass m moves in a horizontal straight line. The only force acting on the particle is a resistance of magnitude $mk(v^2 + c^2)$, where k and c are positive constants and v is the speed. If $v = 2c$ when $t = 0$, find
 - a. The distance travelled and the time taken for the speed to be halved
 - b. The additional distance travelled and the additional time taken for the particle to come to rest
2. A particle of mass m kilograms starts falling from rest having been initially projected vertically upwards from the ground. It experiences air resistance of magnitude mkv^2 on both the upward and downward motion of the journey, where k is a positive constant and v is the velocity of the particle at any instant.
 - a. Show that the terminal velocity, V , of the particle is given by $V = \sqrt{\frac{g}{k}}$
 - b. If W is the velocity of the particle when it hits the ground, show that the distance, D , fallen is given by $D = -\frac{1}{2k} \ln \left(1 - \frac{W^2}{V^2} \right)$
 - c. The maximum height attained by the particle, in terms of its initial velocity, U , and its eventual terminal velocity, V , is given by $\frac{1}{2k} \ln \left(1 + \frac{U^2}{V^2} \right)$. Show that $\frac{1}{W^2} = \frac{1}{U^2} + \frac{1}{V^2}$.
 - d. The particle hits the ground with a velocity of $\frac{3V}{5}$. What initial velocity, U , was needed to achieve this? Given your answer in terms of V .
3. The only force acting on a particle, moving in a straight line, is a resistance $mk(c^2 + v^2)$ acting in that line, where m is the mass of the particle in kilograms, v is the velocity in metres per second and k, c are positive constants. If the distance moved is x metres:
 - a. Show that the acceleration $\ddot{x} = v \frac{dv}{dx}$
 - b. Show that the equation of motion is given by $\frac{dv}{dx} = -\frac{k(c^2 + v^2)}{v}$
 - c. If the particle starts to move with velocity U , show that $x = \frac{1}{2k} \ln \left(\frac{c^2 + U^2}{c^2 + v^2} \right)$
 - d. The particle comes to rest in a distance S metres. Its speed is $\frac{1}{3}U$ when it has moved a distance of $\frac{1}{2}S$ metres.
 - i. Show that $63c^2 = U^2$
 - ii. If the particle has moved a distance of x metres show that $64U^2 e^{-2kx} - 63v^2 = U^2$
 - iii. Show that $0 \leq S \leq \frac{3}{k} \ln 2$
4. A particle of mass m moves in a horizontal straight line. The only force acting on the particle is a resistance of magnitude mkv^3 where v is its speed and k is positive constant. At time t the distance from a fixed point on the line is x . When $t = 0, x = 0$ and $v = u$. Show that $v = \frac{u}{\sqrt{1 + 2ku^2t}} = \frac{u}{1 + kux}$
5. A particle of mass m is set in motion with speed u . Subsequently the only force acting upon the particle directly opposes its motion and is of magnitude $mk(1 + v^2)$ where k is a constant and v is its speed at time t .
 - a. Show that the particle is brought to rest after a time $\frac{1}{k} \tan^{-1} u$
 - b. Find an expression for the distance travelled by the particle in this time.
6. Air resistance to the motion of a particle of mass m has magnitude mkv^2 , where v is the speed of the particle and k is a constant.
 - a. The particle is projected vertically upward under gravity with initial speed v_0 . Write down the equation of motion of the particle during its upward motion and hence show that the greatest height reached is $\frac{1}{2k} \ln \left(\frac{g + kv_0^2}{g} \right)$
 - b. The particle falls from its greatest height. Write down the equation of motion of the particle during its

downward motion and hence find its terminal velocity. If the particle returns to its point of projection with speed v_1 , show that $(g + kv_0^2)(g - kv_1^2) = g^2$

7. A particle moves under gravity in a medium in which the resistance to its motion per unit mass is k times its speed, where k is a constant.

- If the particle falls vertically from rest, show that its terminal velocity is given by $V = \frac{g}{k}$.
- If the particle is projected vertically upward with speed V , show that after time t its speed v and height x are given by $v = V(2e^{-kt} - 1)$ and $x = \frac{1}{k}V(2 - 2e^{-kt} - kt)$. Hence show that the greatest height H that the particle can reach is given by $H = \frac{1}{k}V(1 - \ln 2)$.

8. A projectile is fired vertically upward from the surface of the earth with speed V . The acceleration due to gravity is $\frac{gR^2}{x^2}$ where R is the radius of the earth and x is the distance from the centre of the earth.

- Neglecting air resistance, show that if $V = \sqrt{gR}$, then the speed v of the projectile at distance x from the centre of the earth is given by $v = \sqrt{gR} \sqrt{\frac{2R-x}{x}}$
- Hence show that the projectile reaches a height R above the surface of the earth, and find the time taken to reach this height.

9. The speed v m/s of a point moving along the x -axis is given by $v^2 = 90 - 12x - 6x^3$,

where x m is the displacement of the point from the origin.

- Prove that the motion is simple harmonic.
 - Find the period, the centre of motion, and the amplitude.
10. A particle moves along the x - axis according to the equation $x = 6 \sin 2t - 2\sqrt{3} \cos 2t$, where x is in metres and t is in seconds.
- Express x in the form $R \sin(2t - \alpha)$ where $R > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$.
 - Show that the particle is moving in simple harmonic motion.
 - State the period of this particle's motion.
 - When is the first time that the particle is 6 metres right of the origin?
11. A particle is released from rest at position $\frac{5\pi}{4}$ metres to the right of the origin and travels in a straight line. Its acceleration is described by the equation $\frac{d^2x}{dt^2} = 2 \cos x$ where x is the displacement in metres from the origin O and t is the time in seconds.
- In which direction will the particle first move? Justify your answer.
 - Show that its velocity is given by $v^2 = 4 \sin x + 2\sqrt{2}$
 - Where is the particle stationary? Explain your answer.
 - Is the particle's motion Simple Harmonic? Justify your answer.

Answers:

1. $\frac{1}{2k} \ln \frac{5}{2}, \frac{1}{ck} \tan^{-1} \frac{1}{3}, \frac{1}{2k} \ln 2, \frac{\pi}{4kc}$

2. $U = \frac{3V}{4}$

4.

5. $\frac{1}{2k} \ln(1 + u^2)$

6. $\ddot{x} = -g - kv^2, \dot{x} = g - kv^2, \sqrt{\frac{g}{k}}$

7.

8. $\sqrt{\left(\frac{R}{g}\right) \left(\frac{\pi}{2} + 1\right)}$

9. b. period = $\frac{\sqrt{6}\pi}{3}$, centre of motion = $-1, a = 4$

10. a. $4\sqrt{3} \sin(2t - \frac{\pi}{6})$ c. π d. $\frac{\pi}{4}$

11. a. to the left; c. $x = -\frac{\pi}{4}$; d. No